

- 6 (a) By reference to the photoelectric effect, state what is meant by the work function of a surface.

.....
..... [1]

- (b) Light of wavelength 540 nm is incident on a metal surface having a work function of 2.5 eV.

Determine whether electrons are emitted from the surface.

..... [3]

- (c) Explain whether your conclusion in (b) is affected by the intensity of light incident on the surface.

.....
.....
..... [2]

- (d) Fig. 6.1 below shows some of the energy levels (in electron-volts) of an atom X.

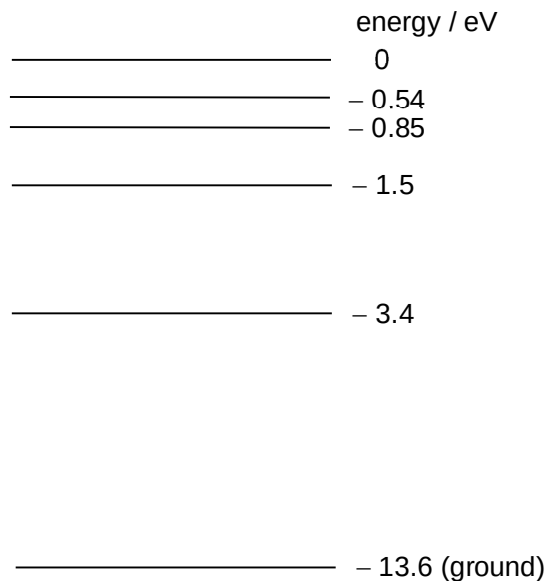


Fig. 6.1

Cool vapour of X at low pressure is bombarded with electrons that are accelerated from rest through an accelerating potential of 12.8 V.

(i) State the ionisation energy of X.

ionisation energy = eV [1]

(ii) Determine the number of different wavelengths of radiation that will be emitted.

number of different wavelengths = [2]

(iii) On Fig. 6.2, draw the spectral lines. Label the line corresponding to the highest energy radiation.



Fig. 6.2

[2]

(iv) Determine the wavelength of the highest energy radiation.

wavelength = m [1]

7 A transformer has 3000 turns in its primary coil. It is used to convert a main supply of root-